

Math 241, F14
Quiz 11

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Evaluate the iterated integral $\int_0^1 \int_0^z \int_0^y 2ze^{-y^2} dx dy dz$.

$$\begin{aligned} & \int_0^1 \int_0^z \int_0^y 2ze^{-y^2} dx dy dz \\ &= \int_0^1 \int_0^z 2ze^{-y^2} [x]_{x=0}^{x=y} dy dz = \int_0^1 \int_0^z 2zye^{-y^2} dy dz \\ &= \int_0^1 z [-e^{-y^2}]_{y=0}^{y=z} dz = \int_0^1 z (-e^{-z^2} + 1) dz \\ &= \left[\frac{e^{-z^2}}{2} + \frac{z^2}{2} \right]_{z=0}^{z=1} = \frac{e^{-1}}{2} \end{aligned}$$

Problem 2. (5 points) Evaluate $\iiint_E 6xy dV$ where E lies under the plane $z = 1 + x + y$ and above the region in the xy -plane enclosed by the curves $y = x$, $y = 0$, and $x = 1$.

$$\begin{aligned} E &= \{(x, y, z) \mid (x, y) \in D, 0 \leq z \leq 1 + x + y\} \\ D &= \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq x\} \\ \Rightarrow \iiint_E 6xy dV &= \iint_D \left[\int_0^{1+x+y} 6xy dz \right] dA \\ &= \int_0^1 \int_0^x \int_0^{1+x+y} 6xy dz dy dx = \int_0^1 \int_0^x 6xy [z]_{z=0}^{z=1+x+y} dy dx \\ &= \int_0^1 \int_0^x 6xy(1+x+y) dy dx = \int_0^1 \int_0^x (6xy + 6x^2y + 6xy^2) dy dx \\ &= \int_0^1 [3xy^2 + 3x^2y^2 + 2xy^3]_{y=0}^{y=x} dx \\ &= \int_0^1 (3x^3 + 3x^4 + 2x^4) dx = \int_0^1 (3x^3 + 5x^4) dx \\ &= \left[\frac{3}{4}x^4 + x^5 \right]_{x=0}^{x=1} = \frac{3}{4} + 1 = \frac{7}{4} \end{aligned}$$

